

Derivation of the Hamiltonian model, Andreev bound states and continuum density of states

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1 Hamiltonian for the Superconductor-Quantum spin Hall insulator-Superconductor diode

1.1 Doppler shift by magnetic field in the spin Hall region

1.1.1 Magnetic Flux and Phase Gradient

In a superconductor, the superconducting phase ϕ_σ of the condensate is related to the vector potential $\mathbf{A}$ via:

$$\mathbf{p}_s = \frac{\hbar}{2} \nabla \phi_\sigma = -\frac{e}{c} \mathbf{A}$$

This comes from the London equation and minimal coupling in superconductivity.

1.1.2 Geometry of the Josephson Junction

From the diagram, we notice the following points:

- The magnetic field $\mathbf{B} = B\hat{z}$ is applied perpendicular to the plane of the QSH junction.
- The superconducting junction has a width W_s along y -direction and a length L_N along the x -direction.

So, the total flux through the QSH junction is:

$$\Phi_\sigma = B \cdot W_s \cdot L_N$$

1.1.3 Gauge Choice and Vector Potential

Choose a Landau gauge:

$$\mathbf{A} = -By\hat{x}$$

Then, the gauge-invariant phase gradient in the x-direction is:

$$\frac{\partial\phi_\sigma}{\partial x} = \frac{2e}{\hbar c}A_x = -\frac{2e}{\hbar c}By$$

The phase difference across a contour enclosing a flux Φ_σ is:

$$\Delta\phi_\sigma = \frac{2\pi\Phi_\sigma}{\Phi_0}$$

So, if you go across a distance L_N in the x -direction, the average phase gradient is:

$$\frac{\partial\phi_\sigma}{\partial x} = \frac{\Delta\phi_\sigma}{L_N} = \frac{2\pi\Phi_\sigma}{\Phi_\sigma L_N}$$

Substitute $\Phi_\sigma = BW_s L_N$:

$$\frac{\partial\phi_\sigma}{\partial x} = \frac{2\pi BW_s L_N}{\Phi_\sigma L_N} = \frac{2\pi BW_s}{\Phi_\sigma}$$

1.1.4 Cooper pair momentum Shift

The supercurrent momentum associated with a superconducting phase gradient is:

$$p_s = \frac{\hbar}{2} \frac{\partial\phi_\sigma}{\partial x}$$

So:

$$p_s = \frac{\hbar}{2} \cdot \frac{2\pi BW_s}{\phi_\sigma} = \pi\hbar \frac{BW_s}{\phi_\sigma},$$

which is the Doppler-shift-induced momentum appearing in the QSH Josephson diode Hamiltonian.

1.1.5 What is Doppler shift in this context?

In a superconductor with a phase gradient, the Cooper pair condensate moves with a finite momentum p_s . When an edge state electron couples to this condensate (say, via Andreev reflection), it feels an energy shift analogous to a Doppler effect:

An electron in a moving medium (the supercurrent) experiences an energy shift depending on its direction of motion.

1.2 Edge modes of a QSH system

On the QSH edge:

1. The dispersion is approximately linear:

$$\epsilon_s(k) = sv_F k$$

2. Here, $s = +1$ for right-movers (spin up), $s = -1$ for left-movers (spin down),
3. v_F is the Fermi velocity of the edge states.

1.3 How does p_s enter?

The superconducting phase gradient in x -direction implies a momentum:

$$p_s = \frac{\hbar}{2} \frac{\partial \phi_\sigma}{\partial x} = \pi \hbar \frac{B W_s}{\phi_\sigma}$$

This momentum gets added to Cooper pairs tunnelling into the edge.

But Cooper pairing is **between time-reversed partners** — so the pair shares p_s between the two electrons (think of it like the centre-of-mass momentum of the pair being p_s). So when a single edge state electron couples to the condensate, it experiences **half** of that momentum shift.

So effectively, the energy of the edge electron shifts by:

$$\delta E = \pm v_F \cdot \frac{p_s}{2}$$

The sign depends on whether it's a right-mover or left-mover. Hence, we write:

$$\delta E = s \frac{v_F p_s}{2}$$

This is a **helicity-dependent shift**, and it's essential for **non-reciprocal superconducting transport** in these systems.

1.4 Proximity induced Cooper pair pairing potential

1.5 Defining the Nambu Spinor

Let's write the Nambu spinor as:

$$\Psi(x) = \begin{pmatrix} \psi_\sigma(x) \\ \psi_\sigma^\dagger(x) \end{pmatrix}$$

In this basis, operators act using the Pauli matrices τ_i where:

1. τ_z : distinguishes particles and holes
2. τ_x, τ_y : create and annihilate Cooper pairs

1.6 Physical Pairing Term

The proximity-induced s-wave pairing couples electrons of opposite spin and momentum (let σ be any quantum number significant for the Hamiltonian system):

$$\hat{H}_\Delta = \Delta(x)e^{i\phi_\sigma(x)}\psi_\sigma^\dagger(x)\psi_{\bar{\sigma}}^\dagger(x) + \text{H.c.}$$

Here:

1. $\Delta(x)$: real, positive pairing amplitude
2. $\phi_\sigma(x)$: spatially varying superconducting phase

The Hermitian conjugate is:

$$\left[\Delta(x)e^{i\phi_\sigma(x)}\psi_\sigma^\dagger(x)\psi_{\bar{\sigma}}^\dagger(x) \right]^\dagger = \Delta(x)e^{i\phi_\sigma(x)}\psi_{\bar{\sigma}}(x)\psi_\sigma(x)$$

So the total term is:

$$\hat{H}_\Delta = \Delta(x)e^{i\phi_\sigma(x)}\psi_\sigma^\dagger\psi_{\bar{\sigma}}^\dagger + \Delta(x)e^{i\phi_\sigma(x)}\psi_{\bar{\sigma}}\psi_\sigma$$

1.7 Represent in Nambu Matrix Form

We now represent this in Nambu space. The general structure of the BdG Hamiltonian is:

$$\hat{H} = \frac{1}{2} \int dx \Psi^\dagger(x) \mathcal{H}(x) \Psi(x)$$

So we are seeking the matrix form of the pairing term $\mathcal{H}_\Delta(x)$, such that:

$$\Psi^\dagger(x) \begin{pmatrix} 0 & \Delta(x)e^{i\phi_\sigma(x)} \\ \Delta(x)e^{i\phi_\sigma(x)} & 0 \end{pmatrix} \Psi(x)$$

This matrix is a 2×2 Hermitian matrix, and we want to express it using Pauli matrices.

1.8 Express Using Pauli Matrices

Recall:

- $\tau_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
- $\tau_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

Now write:

$$\begin{aligned}\Delta(x)e^{i\phi_\sigma(x)} &= \Delta(x)[\cos \phi_\sigma(x) + i \sin \phi_\sigma(x)] \\ \Delta(x)e^{i\phi_\sigma(x)} &= \Delta(x)[\cos \phi_\sigma(x) - i \sin \phi_\sigma(x)]\end{aligned}$$

Thus,

$$\begin{pmatrix} 0 & \Delta(x)e^{i\phi_\sigma(x)} \\ \Delta(x)e^{i\phi_\sigma(x)} & 0 \end{pmatrix} = \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)]$$

So we have:

$$\mathcal{H}_\Delta(x) = \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)]$$

1.9 Method 2: Use Raising and Lowering Operators

The Pauli matrices have ladder operators:

$$\tau_+ = \frac{1}{2}(\tau_x + i\tau_y) \quad \text{and} \quad \tau_- = \frac{1}{2}(\tau_x - i\tau_y)$$

So:

$$\begin{aligned}\Delta(x)e^{i\phi_\sigma(x)}\tau_+ + \Delta(x)e^{i\phi_\sigma(x)}\tau_- &= \Delta(x) \left[\frac{1}{2}(\tau_x + i\tau_y)e^{i\phi_\sigma(x)} + \frac{1}{2}(\tau_x - i\tau_y)e^{i\phi_\sigma(x)} \right] \\ &= \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)]\end{aligned}$$

2 Total Hamiltonian

We study the Bogoliubov–de Gennes (BdG) Hamiltonian for a Superconductor–Quantum Spin Hall Insulator–Superconductor (S–QSHI–S) junction. The Hamiltonian is given for spin $s = \uparrow, \downarrow$ and edge $\sigma = t$ (top), b (bottom) as:

$$\hat{H}_{s,\sigma} = (s\sigma v_F \hat{p}_x - \mu_S)\tau_z + s\frac{v_F p_S}{2} + V_0 h(x)\tau_z + \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)]$$

Where:

- v_F : Fermi velocity
- $\hat{p}_x$: momentum operator
- μ_S : chemical potential
- $V_0 h(x)$: scalar potential profile

- $p_S = \pi\hbar \frac{BW_S}{\phi_\sigma}$: Doppler shift momentum from orbital magnetic field B and width W_S
- $\tau_{x,y,z}$: Pauli matrices in Nambu (particle-hole) space
- $\phi_\sigma(x)$: superconducting phase at edge σ

Why is the kinetic term linear in momentum?

The edge states of a Quantum Spin Hall Insulator (QSHI) are helical, meaning that the spin and direction of motion are locked: spin- $\uparrow$ electrons move in one direction and spin- $\downarrow$ in the opposite. As a result, the dispersion near the Dirac point is linear, and the low-energy Hamiltonian for each helical edge is:

$$H_{\text{edge}} = \pm v_F \hat{p}_x$$

where the sign depends on spin and edge. Therefore, the kinetic term in the BdG Hamiltonian must also be linear in momentum $\hat{p}_x$, as inherited from the helical edge dispersion.

BdG Nambu Spinor

We define the Nambu spinor:

$$\Psi(x) = \begin{pmatrix} \psi_{\uparrow,t}(x) \\ \psi_{\downarrow,b}(x) \\ \psi_{\downarrow,b}^\dagger(x) \\ \psi_{\uparrow,t}^\dagger(x) \end{pmatrix}$$

BdG Hamiltonian Matrix

In this basis, the BdG Hamiltonian becomes a 4×4 matrix:

$$\hat{H}_{\text{BdG}}(x) = \begin{pmatrix} v_F \hat{p}_x - \mu_S + \frac{v_F p_S}{2} + V_0 h(x) & 0 & 0 & \Delta(x) e^{i\phi_{\uparrow,t}(x)} \\ 0 & -v_F \hat{p}_x - \mu_S - \frac{v_F p_S}{2} + V_0 h(x) & \Delta(x) e^{i\phi_{\downarrow,b}(x)} & 0 \\ 0 & \Delta(x) e^{i\phi_{\downarrow,b}(x)} & -v_F \hat{p}_x + \mu_S - \frac{v_F p_S}{2} - V_0 h(x) & 0 \\ \Delta(x) e^{i\phi_{\uparrow,t}(x)} & 0 & 0 & v_F \hat{p}_x + \mu_S + \frac{v_F p_S}{2} - V_0 h(x) \end{pmatrix}$$

This Hamiltonian captures:

- Helical edge modes with spin-momentum locking ($s\sigma v_F \hat{p}_x$ term)
- Doppler shift ($\frac{v_F p_S}{2}$) from the orbital magnetic effect
- Proximity-induced superconductivity via $\Delta(x) e^{\pm i\phi_\sigma(x)}$
- Edge-dependent phases $\phi_{s,t(x)}$ and $\phi_{s,b}(x)$

3 Representation of the wavefunction

3.1 Structure of the total wavefunction

The full Nambu spinor $\Psi_{s,\sigma}(x)$ includes:

1. Particle-hole (Nambu) space, where the BdG Hamiltonian acts. This is represented by $\psi_{s,\sigma}(x)$, a two-component object.
2. Edge space: Encoded by η_σ , eigenvector of σ_z , with eigenvalues $\sigma = \pm 1$.
3. Spin space: Encoded by χ_s , eigenvector of s_z , with eigenvalues $s = \pm 1$.

So, the full Hilbert space is:

$$\mathcal{H} = \mathcal{H}_{\text{Nambu}} \otimes \mathcal{H}_{\text{edge}} \otimes \mathcal{H}_{\text{spin}}$$

and the total spinor $\Psi_{s,\sigma}(x)$ lives in this space.

3.2 Good quantum numbers

Since,

$$[\hat{H}_{BdG}, s_z] = 0 \quad \text{and} \quad [\hat{H}_{BdG}, \sigma_z] = 0$$

This means spin, s and edge index, σ are conserved quantities. Therefore, the BdG Hamiltonian does not mix different spin or edge components. That is:

- If you're in the spin-up ($s = +1$) sector, the Hamiltonian never connects you to spin-down.
- If you're on the top edge ($\sigma = +1$), the Hamiltonian does not mix you with the bottom edge.

So, for each fixed pair (s, σ) , the Hamiltonian reduces to an effective Hamiltonian $\hat{H}_{s,\sigma}$ acting only in Nambu space.

3.3 Effective reduction of the Hilbert space

Given that spin and edge sectors are not mixed, the problem block-diagonalizes into smaller sectors. For a fixed pair (s, σ) , the eigenstate becomes:

$$\Psi_{s,\sigma}(x) = \psi_{s,\sigma}(x) \otimes \eta_\sigma \otimes \chi_s$$

Now, the only part that evolves dynamically under $\hat{H}_{s,\sigma}$ is the Nambu spinor $\psi_{s,\sigma}(x)$, which has two components:

$$\psi_{s,\sigma}(x) = \begin{pmatrix} u_{s,\sigma}(x) \\ v_{s,\sigma}(x) \end{pmatrix}$$

Thus, it is a 2-component spinor: it corresponds to the electron and hole amplitudes in the BdG formalism.

So, although the full wavefunction lives in an 8-dimensional space (2 Nambu $\times$ 2 edge $\times$ 2 spin), conservation of spin and edge quantum numbers lets us project onto a single sector. In each such sector, the Hamiltonian is effectively a 2×2 operator acting on $\psi_{s,\sigma}(x)$, and that's why it becomes a 2-spinor.

3.4 Ansatz for the wavefunction of Andreev Bound State

The Hamiltonian is:

$$\hat{H}_{s,\sigma} = (s\sigma v_F \hat{p}_x - \mu_S) \tau_z + s \frac{v_F p_S}{2} + V_0 h(x) \tau_z + \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)],$$

acting on the Nambu spinor:

$$\psi_{s,\sigma}(x) = \begin{pmatrix} \psi_e \\ \psi_h \end{pmatrix},$$

where τ_x, τ_y, τ_z are Pauli matrices in Nambu space (electron-hole), and $\Delta(x)$, $\phi_\sigma(x)$, and $V_0 h(x)$ define a piecewise inhomogeneous system.

We split the real line into three regions:

1. $x < 0$: Left superconductor (S)
2. $0 < x < L_N$: Normal metal (N)
3. $x > L_N$: Right superconductor (S)

We take

$$\epsilon_s = \epsilon - s v_F p_S / 2,$$

to be the contribution of spin energy added to free electron energy.

3.4.1 Look for bound states

We are interested in Andreev bound states (ABS):

- These states exist inside the superconducting gap, i.e., $|\epsilon_s| < \Delta$.
- This means in the superconducting regions, the solutions are evanescent(decaying), and in the normal region, the solutions are propagating.

3.4.2 General form of wavefunction in each region

Superconducting regions ($x < 0$ and $x > L_N$)

In superconductors:

- The BdG equations admit exponentially decaying solutions for subgap energies.
- The general solution is a combination of an electron-like and a hole-like component with coherence factors.

So the form:

$$\psi_{s,\sigma}(x) \propto \begin{pmatrix} 1 \\ C_{\pm}^{(s)} \end{pmatrix} e^{is\sigma k_F x} e^{\pm \kappa_{\epsilon}^{(s)} x}$$

comes from solving the BdG equation in uniform superconducting regions, assuming a constant pairing potential Δ , where:

$$C_{\pm}^{(s)} = \frac{\epsilon_s \pm i\sqrt{\Delta^2 - \epsilon_s^2}}{\Delta}, \quad \kappa_{\epsilon}^{(s)} = \frac{\sqrt{\Delta^2 - \epsilon_s^2}}{\hbar v_F}$$

These factors are standard for evanescent solutions of the BdG equations with energy ϵ_s less than Δ .

The pre-factors $a_{s,\sigma}$ and $b_{s,\sigma}$ are coefficients to be determined by boundary matching.

Also note: $e^{i\phi_{\sigma}}$ in the right superconductor accounts for the superconducting phase difference.

Normal region $0 < x < L_N$

Here:

- $\Delta(x) = 0$, so the BdG equation reduces to two uncoupled modes:

1. One for electrons
2. One for holes

- Solutions are plane waves, since no pairing means free propagation.

Thus, the general form is a superposition of an electron-like wave and a hole-like wave:

$$\psi_{s,\sigma}(x) = e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{is\sigma(k_N + k_{\epsilon}^{(s)})x} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{is\sigma(k_N - k_{\epsilon}^{(s)})x}$$

where:

1. $k_N = \mu_N / \hbar v_F$
2. $k_{\epsilon}^{(s)} = \epsilon_s / \hbar v_F$

Defining all parameters properly

Let's make sure we're tracking all symbols:

1. $s = \pm 1$: chirality (direction)
2. $\sigma = \pm 1$: spin
3. $\epsilon_s = \epsilon - sv_F p_S / 2$: spin- and chirality-shifted energy
4. $\Delta(x)$: real superconducting gap profile
5. $\phi_{\sigma}(x) = \phi_{\sigma} + \sigma \frac{p_S L_N}{\hbar}$: superconducting phase (spin-dependent), ϕ_{σ} is the resulting in a global phase difference

6. $\kappa_\epsilon^{(s)}$: decay wavevector in superconducting region for evanescent modes, wavevector in normal region
7. $C_\pm^{(s)}$: superconducting coherence factors
8. $L_{\text{tot}} = L_N + L_S$: total effective length for normalization

Summary of ansatz choice

For ABS, that is, states with $|\epsilon_s| < \Delta$ and thus states localized around the N region, we can make the piecewise ansatz

$$\psi_{s,\sigma}(x) = \frac{1}{\sqrt{L_{\text{tot}}}} \begin{cases} a_{s,\sigma} \begin{pmatrix} 1 \\ C_{s\sigma}^{(s)} \end{pmatrix} e^{is\sigma k_F x} e^{\kappa_\epsilon^{(s)} x}, x < 0, \\ e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{is\sigma(k_N + k_\epsilon^{(s)})x} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{is\sigma(k_N - k_\epsilon^{(s)})x}, 0 < x < L_N, \\ b_{s,\sigma} \begin{pmatrix} 1 \\ C_{-s\sigma}^{(s)} e^{i\phi_\sigma} \end{pmatrix} e^{is\sigma k_F x} e^{-\kappa_\epsilon^{(s)} x}, x > L_N \end{cases}$$

4 Applying the δ - potential to find out the ABS energies

To derive the matching condition across a delta barrier for the BdG-like Hamiltonian:

$$\hat{H}_{s,\sigma} = (s\sigma v_F \hat{p}_x - \mu_S) \tau_z + s \frac{v_{FPS}}{2} + V_0 h(x) \tau_z + \Delta(x) [\tau_x \cos \phi_\sigma(x) - \tau_y \sin \phi_\sigma(x)],$$

we focus on the effect of the delta-function barrier at $x = 0$, where:

- $h(x) = L_N \delta(x)$ is a Dirac delta.
- The potential term becomes $V_0 L_N \delta(x) \tau_z$.

4.1 Schrodinger Equation

We begin with the BdG equation:

$$\hat{H}_{s,\sigma} \psi_{s,\sigma}(x) = E \psi_{s,\sigma}(x).$$

We now integrate both sides of this equation across an infinitesimally small region around $x = 0$, say from $-\eta$ to $+\eta$, and let $\eta \rightarrow 0^+$.

4.2 Integration Across the Delta Barrier

We only need to consider terms in the Hamiltonian that involve derivatives or singularities. The derivative term in the Hamiltonian is:

$$\hat{p}_x = -i\hbar\partial_x.$$

The delta-function appears in:

$$V_0\delta(x)\tau_z.$$

So, we integrate:

$$\int_{-\eta}^{\eta} \hat{H}_{s,\sigma}\psi(x) dx = \int_{-\eta}^{\eta} E\psi(x) dx.$$

Since $\psi(x)$ is continuous on both sides of the junction except possibly at $x = 0$, and all smooth terms vanish in the limit $\eta \rightarrow 0$, only the discontinuity-producing terms contribute. These are:

1. The derivative term $s\sigma v_F \hat{p}_x \tau_z = -is\sigma \hbar v_F \tau_z \partial_x$.
2. The delta term $V_0 L_N \delta(x) \tau_z$.

So we integrate:

$$\int_{-\eta}^{\eta} [-is\sigma \hbar v_F \tau_z \partial_x + V_0 L_N \delta(x) \tau_z] \psi(x) dx.$$

The first term gives:

$$-is\sigma \hbar v_F \tau_z [\psi(x)]_{-\eta}^{+\eta} = -is\sigma \hbar v_F \tau_z [\psi(0^+) - \psi(0^-)],$$

and the second term gives:

$$V_0 L_N \tau_z \psi(0).$$

But since the delta function samples the value at $x = 0$, and there may be a discontinuity, we usually take the average:

$$\delta(x)\psi(x) \longrightarrow \delta(x) \frac{1}{2} [\psi(0^+) + \psi(0^-)].$$

So the full integrated equation becomes:

$$-is\sigma \hbar v_F \tau_z [\psi(0^+) - \psi(0^-)] + V_0 L_N \tau_z \cdot \frac{1}{2} [\psi(0^+) + \psi(0^-)] = 0.$$

4.3 Solve for the Matching Condition

Rewriting:

$$-is\sigma\hbar v_F [\psi(0^+) - \psi(0^-)] + \frac{V_0 L_N}{2} [\psi(0^+) + \psi(0^-)] = 0.$$

Rearranging terms:

$$\left(-is\sigma\hbar v_F + \frac{V_0 L_N}{2}\right) \psi(0^+) = \left(-is\sigma\hbar v_F - \frac{V_0 L_N}{2}\right) \psi(0^-).$$

Divide both sides by $-is\sigma\hbar v_F$, and define:

$$Z_0 = \frac{V_0 L_N}{\hbar v_F},$$

then:

$$\left(1 - i \frac{Z_0}{2s\sigma}\right) \psi(0^+) = \left(1 + i \frac{Z_0}{2s\sigma}\right) \psi(0^-).$$

This has the form:

$$\psi(0^+) = \left(\frac{1 + i \frac{Z_0}{2s\sigma}}{1 - i \frac{Z_0}{2s\sigma}}\right) \psi(0^-).$$

The right-hand side is a pure phase, since:

$$\left|\frac{1 + i\alpha}{1 - i\alpha}\right| = 1 \quad \text{for real } \alpha.$$

Let's define:

$$e^{-is\sigma Z_0} := \frac{1 + i \frac{Z_0}{2s\sigma}}{1 - i \frac{Z_0}{2s\sigma}},$$

which is approximately valid for small Z_0 . But this can also be justified as the exact matching phase condition from the delta-function potential. To show this let us define:

$$\alpha := \frac{Z_0}{2s\sigma},$$

so the factor becomes:

$$\frac{1 + i\alpha}{1 - i\alpha}.$$

We can write:

$$\frac{1 + i\alpha}{1 - i\alpha} = \frac{(1 + i\alpha)(1 + i\alpha)}{(1 - i\alpha)(1 + i\alpha)} = \frac{(1 + i\alpha)^2}{1 + \alpha^2}.$$

So,

$$\frac{1 + i\alpha}{1 - i\alpha} = \frac{1 - \alpha^2 + 2i\alpha}{1 + \alpha^2}.$$

Thus, this is a complex number with modulus 1:

$$\left| \frac{1+i\alpha}{1-i\alpha} \right| = 1.$$

Any complex number with modulus 1 can be written as $e^{i\theta}$ for some real θ . Expanding logarithmically for small Z_0 and taking the logarithm of both sides:

$$\ln \left(\frac{1+i\alpha}{1-i\alpha} \right) = \ln(1+i\alpha) - \ln(1-i\alpha).$$

Using the Taylor expansion:

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

So for small α ,

$$\ln(1+i\alpha) = i\alpha - \frac{(i\alpha)^2}{2} + \dots = i\alpha + \frac{\alpha^2}{2} + \dots,$$

$$\ln(1-i\alpha) = -i\alpha - \frac{\alpha^2}{2} + \dots.$$

Then:

$$\ln \left(\frac{1+i\alpha}{1-i\alpha} \right) = \left(i\alpha + \frac{\alpha^2}{2} \right) - \left(-i\alpha - \frac{\alpha^2}{2} \right) = 2i\alpha + \alpha^2.$$

So,

$$\frac{1+i\alpha}{1-i\alpha} \approx \exp(2i\alpha) = \exp \left(i \frac{Z_0}{s\sigma} \right).$$

Therefore, we have:

$$\psi(0^+) = e^{i \frac{Z_0}{s\sigma}} \psi(0^-).$$

Consistent because:

$$\frac{Z_0}{s\sigma} = s\sigma Z_0 \quad (\text{since } s^2 = \sigma^2 = 1).$$

So the identity:

$$\frac{1 + i \frac{Z_0}{2s\sigma}}{1 - i \frac{Z_0}{2s\sigma}} = e^{-is\sigma Z_0}$$

is exact to leading order in small Z_0 , and gives the correct phase accumulated by the wavefunction across the delta barrier.

This justifies writing the exact matching condition as:

$$\boxed{\psi(0^+) = e^{-is\sigma Z_0} \psi(0^-)}$$

whenever $Z_0 \ll 1$, and it holds up to terms of order $\mathcal{O}(Z_0^2)$, which are negligible in the weak barrier limit.

4.3.1 Final Result

Thus, the matching condition across the delta barrier is:

$$\boxed{\psi_{s,\sigma}(0^+) = e^{-is\sigma Z_0} \psi_{s,\sigma}(0^-)}$$

This phase shift captures the effect of the potential barrier, and allows solving for Andreev bound states (ABS) using wavefunction matching.

4.4 Finding the ABS energies from boundary condition matching

4.4.1 Evaluating the ansatz on both sides of $x = 0$

The ansatz is:

$$\psi_{s,\sigma}(x) = \frac{1}{\sqrt{L_{\text{tot}}}} \begin{cases} a_{s,\sigma} \begin{pmatrix} 1 \\ C_{s\sigma}^{(s)} \end{pmatrix} e^{is\sigma k_F x} e^{\kappa_\epsilon^{(s)} x}, & x < 0 \\ e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{is\sigma(k_N + k_\epsilon^{(s)})x} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{is\sigma(k_N - k_\epsilon^{(s)})x}, & 0 < x < L_N \\ b_{s,\sigma} \begin{pmatrix} 1 \\ C_{-s\sigma}^{(s)} e^{i\phi_\sigma} \end{pmatrix} e^{is\sigma k_F x} e^{-\kappa_\epsilon^{(s)} x}, & x > L_N \end{cases}$$

At $x = 0^-$ (In superconductor just before the interface):

$$\psi_{s,\sigma}(0^-) = \frac{a_{s,\sigma}}{\sqrt{L_{\text{tot}}}} \begin{pmatrix} 1 \\ C_{s\sigma}^{(s)} \end{pmatrix}$$

At $x = 0^+$ (just after the interface in the normal region):

$$\psi_{s,\sigma}(0^+) = \frac{1}{\sqrt{L_{\text{tot}}}} \left[e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right]$$

Thus the phase matching condition becomes:

$$\begin{pmatrix} e_{s,\sigma} \\ h_{s,\sigma} \end{pmatrix} = e^{-is\sigma Z_0} a_{s,\sigma} \begin{pmatrix} 1 \\ C_{s\sigma}^{(s)} \end{pmatrix}$$

4.4.2 Express $e_{s,\sigma}$, $h_{s,\sigma}$ in terms of $a_{s,\sigma}$

From above:

1. Upper component:

$$e_{s,\sigma} = e^{-is\sigma Z_0} a_{s,\sigma},$$

2. Lower component:

$$h_{s,\sigma} = e^{-is\sigma Z_0} a_{s,\sigma} C_{s\sigma}^{(s)}$$

Now, let's use the boundary condition at $x = L_N$ to link $a_{s,\sigma}$ and $b_{s,\sigma}$, and extract the quantization.

4.4.3 Applying the phase matching at $x = L_N$

Next, we match at $x = L_N$ using the condition:

$$\psi_{s,\sigma}(L_N^+) = e^{-is\sigma Z_0} \psi_{s,\sigma}(L_N^-)$$

Left of L_N (normal region):

$$\psi_{s,\sigma}(L_N^-) = \frac{1}{\sqrt{L_{\text{tot}}}} \left[e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{is\sigma(k_N + k_\epsilon^{(s)})L_N} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{is\sigma(k_N - k_\epsilon^{(s)})L_N} \right]$$

Right of L_N (superconductor):

$$\psi_{s,\sigma}(L_N^+) = \frac{1}{\sqrt{L_{\text{tot}}}} b_{s,\sigma} \begin{pmatrix} 1 \\ C_{-s\sigma}^{(s)} e^{i\phi_\sigma} \end{pmatrix} e^{is\sigma k_F L_N} e^{-\kappa_\epsilon^{(s)} L_N}$$

Using the boundary condition:

$$b_{s,\sigma} \begin{pmatrix} 1 \\ C_{-s\sigma}^{(s)} e^{i\phi_\sigma} \end{pmatrix} e^{is\sigma k_F L_N} e^{-\kappa_\epsilon^{(s)} L_N} = e^{-is\sigma Z_0} \left[e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{is\sigma(k_N + k_\epsilon^{(s)})L_N} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{is\sigma(k_N - k_\epsilon^{(s)})L_N} \right]$$

Substituting $e_{s,\sigma}$ and $h_{s,\sigma}$ from earlier:

$$b_{s,\sigma} \begin{pmatrix} 1 \\ C_{-s\sigma}^{(s)} e^{i\phi_\sigma} \end{pmatrix} e^{is\sigma k_F L_N} e^{-\kappa_\epsilon^{(s)} L_N} = a_{s,\sigma} e^{-2is\sigma Z_0} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{is\sigma(k_N + k_\epsilon^{(s)})L_N} + C_{s\sigma}^{(s)} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{is\sigma(k_N - k_\epsilon^{(s)})L_N} \right]$$

Matching each component:

1st component:

$$b_{s,\sigma} e^{is\sigma k_F L_N} e^{-\kappa_\epsilon^{(s)} L_N} = a_{s,\sigma} e^{-2is\sigma Z_0} e^{is\sigma(k_N + k_\epsilon^{(s)})L_N}$$

2nd component:

$$b_{s,\sigma} C_{-s\sigma}^{(s)} e^{i\phi_\sigma} e^{is\sigma k_F L_N} e^{-\kappa_\epsilon^{(s)} L_N} = a_{s,\sigma} C_{s\sigma}^{(s)} e^{-2is\sigma Z_0} e^{is\sigma(k_N - k_\epsilon^{(s)})L_N}$$

Divide the second equation by the first:

$$\frac{C_{-s\sigma}^{(s)} e^{i\phi_\sigma}}{1} = \frac{C_{s\sigma}^{(s)} e^{-2is\sigma k_\epsilon^{(s)} L_N}}{1}$$

This gives:

$$C_{-s\sigma}^{(s)} = C_{s\sigma}^{(s)} e^{-2is\sigma k_\epsilon^{(s)} L_N} e^{i\phi_\sigma}$$

Using the definition:

$$C_{\pm}^{(s)} = \frac{\epsilon_s \pm i\sqrt{\Delta^2 - \epsilon_s^2}}{\Delta} = e^{\pm i \arccos(\epsilon_s/\Delta)} \Rightarrow C_{-s\sigma}^{(s)} = (C_{s\sigma}^{(s)})^*$$

Hence:

$$e^{-2is\sigma k_\epsilon^{(s)} L_N} e^{i\phi_\sigma} = \frac{C_{-s\sigma}^{(s)}}{C_{s\sigma}^{(s)}} = e^{-2i \arccos(\epsilon_s/\Delta)}$$

Taking argument on both sides:

$$-2s\sigma \frac{\epsilon_s L_N}{\hbar v_F} + \phi_\sigma = -2 \arccos(\epsilon_s/\Delta) + 2\pi n \Rightarrow \lim_{L_N \rightarrow 0} \arccos(\epsilon_s/\Delta) = -s\sigma \frac{\phi_\sigma}{2} + \pi n$$

This is the quantization condition for Andreev bound states in an S/N/S edge with a δ -like barrier of strength Z_0 , incorporating the phase shift due to infinitesimal normal region length and superconducting phase difference.

5 Continuum density of states

5.1 Krein-Friedel relation

Let us derive the Krein–Friedel relation (the Friedel sum rule). This beautiful result connects scattering theory with the density of states (DOS) and is widely used in mesoscopic and condensed matter physics.

Suppose we have a 1D or spherically symmetric scattering problem, where the asymptotic wavefunction is modified by a scattering phase shift $\delta_l(\epsilon)$ in partial wave channel l .

Each partial wave contributes to the density of states as:

$$\Delta\rho_l(\epsilon) = \frac{1}{\pi} \frac{d\delta_l(\epsilon)}{d\epsilon}$$

Summing over all channels:

$$\Delta\rho(\epsilon) = \sum_l \Delta\rho_l(\epsilon) = \frac{1}{\pi} \sum_l \frac{d\delta_l}{d\epsilon}$$

Now recall that the scattering matrix in each channel is:

$$S_l(\epsilon) = e^{2i\delta_l(\epsilon)} \Rightarrow \delta_l = \frac{1}{2i} \ln S_l$$

So:

$$\frac{d\delta_l}{d\epsilon} = \frac{1}{2i} \frac{d}{d\epsilon} \ln S_l(\epsilon)$$

Summing over l :

$$\boxed{\Delta\rho(\epsilon) = \frac{1}{2\pi i} \frac{d}{d\epsilon} \ln \det S(\epsilon)}$$

This relates the change in the total density of states due to the presence of a scattering potential to the energy derivative of the log-determinant of the scattering matrix.

5.1.1 Physical Meaning

1. A rapid change in the scattering phase means that a state is being "pushed" into or out of the spectrum — a resonance or bound state.
2. This change shows up as a local increase in DOS.
3. The total number of extra states (integrated $\Delta\rho$) gives the number of bound states.

The physical and mathematical structure of this section, "Continuum States," discusses the scattering states (as opposed to bound Andreev states) in a superconducting-normal-superconducting (SNS) Josephson junction, especially in the regime where the quasiparticle energy $\epsilon_s > \Delta$, i.e., in the continuum above the superconducting gap.

5.2 Scattering matrix approach

5.2.1 Physical Context and meaning of scattering states

In the previous section, the system focused on subgap bound states—i.e., discrete Andreev levels formed due to coherent multiple Andreev reflections inside a weak link (N region).

Here, instead, we study the continuum quasiparticles, i.e., those with energy above the superconducting gap, $\epsilon_s > \Delta$, for which:

- The wavefunctions are propagating instead of evanescent.
- There is no bound state, but these states still encode useful information like phase shifts.

5.2.2 Ansatz for the Wavefunction $\psi_{s,\sigma}(x)$:

For a right-moving, electron-like quasiparticle, the full spatial wavefunction is broken into three regions:

(i) $x < 0$ —i.e., Left superconductor:

Incoming and reflected waves (from the NS interface):

$$\psi_{s,\sigma}(x) = \frac{1}{\sqrt{L_{\text{tot}}}} \left[\begin{pmatrix} u_s \\ v_s \end{pmatrix} e^{i(k_F + q_e^{(s)})x} + r_{eh}^{s,\sigma} \begin{pmatrix} v_s \\ u_s \end{pmatrix} e^{i(k_F - q_e^{(s)})x} \right]$$

1. u_s, v_s are BCS coherence factors (depends on ϵ_s , see below).
2. The first term is an incoming electron-like quasiparticle.
3. The second term is an Andreev-reflected hole (with amplitude $r_{eh}^{s,\sigma}$).

(ii) $0 < x < L_N$ — Normal region:

Two independent waves:

$$e_{s,\sigma} \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{i(k_N + k_\epsilon^{(s)})x} + h_{s,\sigma} \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{i(k_N - k_\epsilon^{(s)})x}$$

These are electron and hole components in the normal metal.

(iii) $x > L_N$ — Right superconductor:

Transmitted wave only:

$$t_{ee}^{s,\sigma} \begin{pmatrix} u_s \\ v_s e^{-i\varphi_\sigma} \end{pmatrix} e^{i(k_F + q_\epsilon^{(s)})x}$$

The phase factor φ_σ encodes the superconducting phase difference across the junction, which can be spin-dependent.

5.2.3 Coherence Factors and Momentum Shifts:

The BCS coherence factors are defined as:

$$u_s^2 = \frac{1}{2} \left(1 + \frac{\sqrt{\epsilon_s^2 - \Delta^2}}{\epsilon_s} \right), \quad v_s^2 = 1 - u_s^2$$

The wavevectors are:

- $q_\epsilon^{(s)} = \frac{\sqrt{\epsilon_s^2 - \Delta^2}}{\hbar v_F}$ — the excitation momentum inside the superconductor.
- $k_\epsilon^{(s)} = \epsilon_s / \hbar v_F$ wavevector inside the normal region.

5.2.4 S-Matrix Structure:

The scattering matrix S_{SNS} is constructed from the reflection and transmission coefficients for quasiparticles across the junction.

Due to the helical edge states (or spin-momentum locking) in a topological system, the S-matrix decouples into top and bottom edge contributions, each labelled by S_{SNS}^t and S_{SNS}^b .

These matrices are written explicitly in terms of scattering amplitudes for each spin channel and quasiparticle type (electron/hole). These states then allow us to set up the S matrix as

$$S_{\text{SNS}} = \begin{pmatrix} S_{\text{SNS}}^t & 0 \\ 0 & S_{\text{SNS}}^b \end{pmatrix}$$

with

$$S_{\text{SNS}}^t = \begin{pmatrix} t_{ee}^{\uparrow,t} & 0 & 0 & r_{he}^{\uparrow,t} \\ 0 & t_{hh}^{\downarrow,t} & r_{eh}^{\downarrow,t} & 0 \\ 0 & r_{he}^{\downarrow,t} & t_{ee}^{\downarrow,t} & 0 \\ r_{eh}^{\uparrow,t} & 0 & 0 & t_{hh}^{\uparrow,t} \end{pmatrix}$$

and

$$S_{\text{SNS}}^b = \begin{pmatrix} t_{ee}^{\downarrow,b} & 0 & 0 & r_{he}^{\downarrow,b} \\ 0 & t_{hh}^{\uparrow,b} & r_{eh}^{\uparrow,b} & 0 \\ 0 & r_{he}^{\uparrow,b} & t_{ee}^{\uparrow,b} & 0 \\ r_{eh}^{\downarrow,b} & 0 & 0 & t_{hh}^{\downarrow,b} \end{pmatrix}.$$

The scattering matrix $S_{\text{SNS}}^\sigma(\epsilon, \phi)$ allows us then to compute the (ϕ -dependent part of the) continuum DOS of the Josephson junction.

5.2.5 Continuum DOS from the S-matrix:

The density of states (DOS) $\rho_0^\sigma(\epsilon, \phi)$ due to the continuum is obtained via the Friedel relation:

$$\rho_0^\sigma(\epsilon, \phi) = \frac{1}{2\pi i} \frac{\partial}{\partial \epsilon} \ln \det S_{\text{SNS}}^\sigma$$

This is essentially the energy derivative of the scattering phase shift and is the continuum analogue of the Friedel sum rule.

Because the total scattering matrix is block-diagonal, the determinant factorizes:

$$\det S_{\text{SNS}} = \det S_{\text{SNS}}^t \cdot \det S_{\text{SNS}}^b$$

5.2.6 Analytical Expression for $\det S_{\text{SNS}}^t$:

For the δ -function junction model, an analytical expression is derived for $\det S_{\text{SNS}}^t$, capturing its energy and phase dependence.

It is got from By applying the boundary conditions:

1. $\psi_{s,\sigma}(0^+) = e^{-is\sigma Z_0} \psi_{s,\sigma}(0^-)$
2. $\psi_{s,\sigma}(L_N^+) = e^{-is\sigma Z_0} \psi_{s,\sigma}(L_N^-)$
3. $\arccos(\epsilon_s/\Delta) = -s\sigma \frac{\phi_\sigma}{2} + \pi n$

to the piecewise-defined wavefunction, we find the reflection and transmission amplitudes as:

$$r_{eh}^{s,\sigma} = \frac{u_s v_s (e^{i\varphi_\sigma} - e^{2iL_N k_\epsilon})}{-u_s^2 e^{i\varphi_\sigma} + v_s^2 e^{2iL_N k_\epsilon}}$$

and,

$$t_{ee}^{s,\sigma} = \frac{(u_s^2 - v_s^2) e^{-i[L_N(k_F - k_N - k_\epsilon + q_\epsilon) + 2Z_0 s\sigma - \varphi_\sigma]}}{u_s^2 e^{i\varphi_\sigma} - v_s^2 e^{2iL_N k_\epsilon}}.$$

Substituting these and calculating the determinant,

$$\det S_{\text{SNS}}^t = \prod_{s=\uparrow,\downarrow} \frac{\epsilon_s^2 - \Delta^2 \cos^2(\phi_t/2)}{\epsilon_s^2 \cos \phi_t - \Delta^2 \cos^2(\phi_t/2) + is\epsilon_s \sqrt{\epsilon_s^2 - \Delta^2} \sin \phi_t}$$

This form emphasizes how the phase difference and quasiparticle energy control the density of states, and it also reflects how Andreev reflection phase shifts appear in the scattering spectrum.

5.3 Analytical expression for DOS

The analytical expression for the DOS in continuum states was calculated and is given by the Krein-Friedel relation:

$$\rho_s^\sigma(\epsilon, \phi^\sigma) = \frac{s\sigma\Delta^2}{2\pi} \frac{\Theta(\epsilon_s^2 - \Delta^2)\text{sgn}(\epsilon_s)\sin\phi^\sigma}{\sqrt{\epsilon_s^2 - \Delta^2(\epsilon_s^2 - \Delta^2\cos^2\frac{\phi^\sigma}{2})}}$$

5.3.1 Summary of Concepts Involved:

- Continuum quasiparticles are described by scattering theory, not bound state quantization.
- The S-matrix fully encodes the energy dependence of scattering amplitudes.
- The continuum density of states is derived from the energy derivative of the S-matrix determinant.
- The phase sensitivity of the continuum DOS reflects Josephson's physics and is crucial for computing things like current noise, heat currents, or dissipative responses in SNS junctions.